

Paediatric drug optimization for dengue

Meeting report, 23 October 2025



**World Health
Organization**

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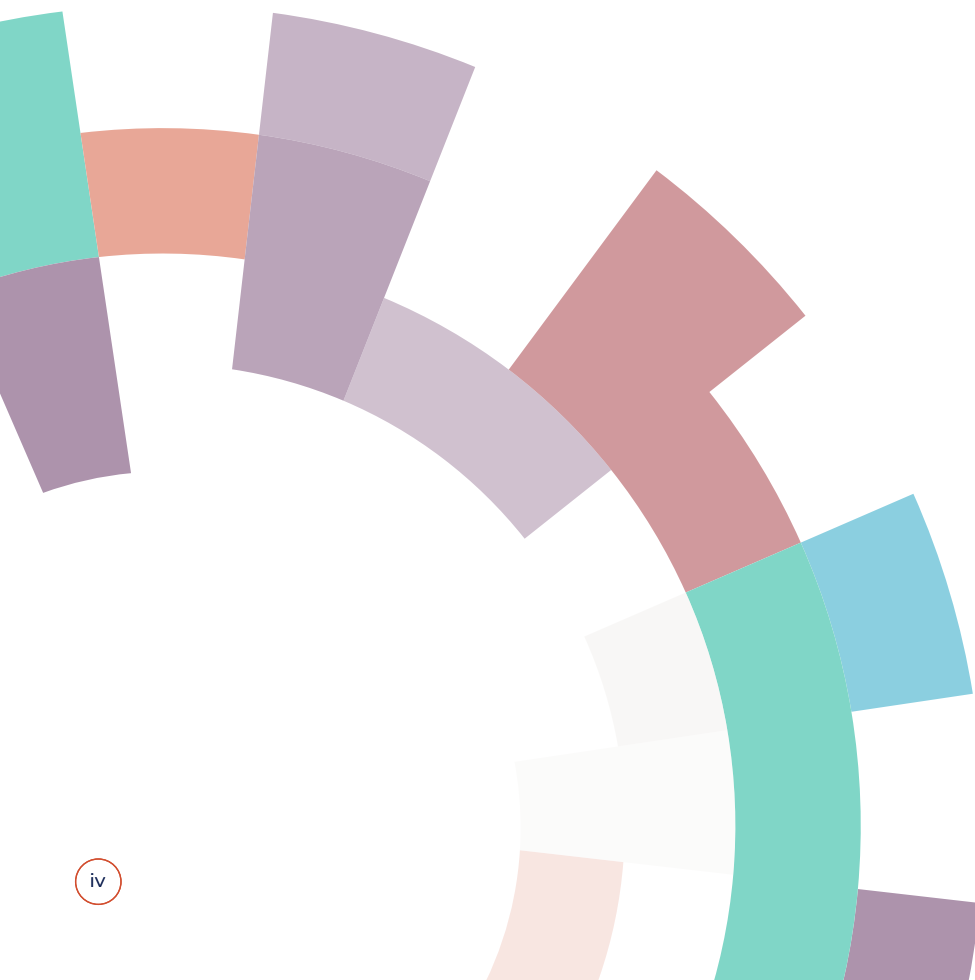
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The need for paediatric drug optimization

The development of medicines for children lags unacceptably behind that for adults by nearly a decade.

Following the resolution at the Sixty-ninth World Health Assembly on promoting innovation and access to quality, safe, efficacious and affordable medicines for children ([1](#)), WHO and partners have increased their efforts to deliver on this global commitment and have scaled up activities to ensure that age-appropriate formulations are available for children. The Global Accelerator for Paediatric Formulations Network (GAP-f), a WHO-hosted network, works across the life cycle of drug development to accelerate the investigation, development and introduction of optimal formulations for children ([2](#)). Priority-setting is the first step to enable a targeted approach to research and development. Developing a priority drug portfolio of the most needed formulations for children is essential to streamline researchers' and supplier's efforts and resources around specific dosage forms and formulations that address the most urgent needs for children. This is especially important since the market for medicines for children is often small and/or fragmented, resulting in limited volumes with potential market failures.

Paediatric drug optimization (PADO) exercises to identify key priority products and their preferred product characteristics for research and development have been successfully undertaken for HIV, hepatitis C, tuberculosis, coronavirus disease 2019 (COVID-19), antibiotics, neglected tropical diseases, childhood cancer, respiratory syncytial virus, malaria and epilepsy, demonstrating their potential and impact to accelerate access to optimal formulations in the context of fragmented, small markets for medicines for children. To provide further guidance to support similar processes for optimizing drugs for children in other disease areas, GAP-f has published a guidance document to undertake a PADO process and adapt it to the specific needs of each disease area ([3](#)).

PADO processes are not guideline processes and as such are not intended to endorse the use of products that have not been fully assessed by a WHO guideline development group. However, priority-setting provides a clear signal that specific formulations and products are of interest in the short, medium or long term and that stakeholders should work together towards completing the investigation and development of age-appropriate formulations.

PADO for dengue

Dengue is a mosquito-borne viral disease of expanding geographical range and increasing incidence (4).

WHO named dengue one of the top 10 threats to global health in 2019 following a 10-fold increase in reported cases from 500 000 to 5.2 million between 2000 and 2019 driven by climate change and globalization (5). A further surge and geographical expansion in the wake of the COVID-19 pandemic saw dengue being graded the highest level of emergency under the WHO Emergency Response Framework in 2023. More than 14 million dengue cases and more than 10 000 dengue-related deaths were reported in 2024, about twice the 6.5 million cases and more than 6800 deaths reported in 2023 (6). Dengue is now endemic in more than 100 countries, and half the world living in tropical and subtropical regions is at risk (4). The vast majority of people with dengue are children younger than 15 years. Young children are also at higher risk of more severe outcomes and complications from dengue (7).

Dengue causes a spectrum of illness from mild fever to severe disease with plasma leakage and shock. Infants and children with secondary heterologous dengue infections are most at risk for severe dengue disease.

There are no licensed treatments for dengue fever, and supportive care remains the mainstay of clinical management as detailed in the *WHO guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever* (8). The WHO Global Strategic Preparedness, Readiness and Response Plan to tackle dengue and other *Aedes*-borne arboviruses highlighted the need for safe and effective medicines to prevent progression to severe disease, organ failure and death (5). Several drug candidates (small molecules and monoclonal antibodies) are now in Phase 2 trials for dengue treatment. WHO has therefore developed target product profiles (TPPs) for treatments for non-severe and severe dengue to support the ongoing clinical development of dengue therapeutics approaching Phase 3 pivotal trials (*WHO target product profiles, preferred product characteristics, and target regimen profiles: standard procedure*, second edition, unpublished internal document) (9).

Objectives

The PADO for dengue exercise aimed to guide paediatric-focused research and development by bringing a paediatric-specific perspective to the TPP development process and identifying priority areas for advancing dengue therapeutics for children. The specific objectives of PADO for dengue included:

- to discuss paediatric-specific considerations to inform the ongoing development of the TPP for dengue therapeutics;
- to review ongoing studies investigating new dengue therapeutics for adults and children and reach consensus on the most promising candidates to give priority to for paediatric investigation in the short term (PADO priority list; 3–5 years) and in the medium to long term (PADO watch list; 5–10 years); and

- to review ongoing research efforts on treating children with dengue and to identify and agree on the critical research gaps to be addressed to inform the approval and optimal use of novel therapeutics for children with dengue.

The PADO for dengue exercise enables alignment across funders, procurers, market-coordination entities, researchers, innovators, generics manufacturers, product development partnerships and regulators, ensuring that the specific needs of children with dengue are considered upfront in research and development programmes and that children can benefit from innovations as soon as they become available.

Methods and meeting proceedings

Available WHO resources on dengue were reviewed, including the *WHO guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever* published in July 2025 ([8](#)) and the draft TPP for dengue therapeutics released for public consultation in June 2025 ([10](#)). The feedback received through the public consultation on the TPP was reviewed and incorporated into the discussions on the paediatric considerations for dengue therapeutics. The priority-setting exercise focused on candidate antiviral agents and monoclonal antibodies in Phase 2 and 3

clinical development, drawing on a pipeline landscape developed by the Drugs for Neglected Diseases initiative. In preparation for the meeting, a dedicated priority-setting framework (Table 1) was developed and detailed with information extracted from WHO publications, scientific publications, clinical trial registries and targeted outreach to relevant developers. The framework provided a structured basis for discussions aimed at identifying therapeutics to be given priority for research and development for children with dengue.

The PADO for dengue meeting was held virtually on 23 October 2025 ([Annex 1](#)) and brought together academics, researchers, clinical experts, regulators, funders and other key stakeholders involved in research and development related to dengue ([Annex 2](#)). Conflict-of-interest declarations were collected for all participants and closely reviewed. None of the participants declared conflicts of interest. United Nations agencies (United Nations Children’s Fund) and GAP-f partners (European

Paediatric Formulation Initiative, Médecins Sans Frontières, Drugs for Neglected Diseases initiative, Medicines Patent Pool and Penta) attended the meeting in an observer capacity.

The consensus on the paediatric aspects of the TPP for dengue therapeutics and on the priority therapeutics to be further investigated and/or developed for infants and children was reached through guided discussions informed by the prepopulated framework described above.

Table 1. Attributes included in the priority-setting frameworks of PADO for dengue for monoclonal antibodies and antiviral agents

Mechanism of action	Drug–drug interactions
Current stage of development	Activity against difficult-to-treat resistance profiles and resistance potential
Efficacy	Laboratory and clinical monitoring
Safety and tolerability	Screening required before administration
Route of administration	Already used for children with another indication
Tolerability profile and use for people with organ impairment	Paediatric formulation
Side-effects	Storage requirements for marketed or study medicine
Preclinical data and signals	Regulatory aspects

Summary of discussions

Ensuring that the TPPs for dengue therapeutics address the needs of children

The discussion focused on the TPP characteristics requiring a specifically paediatric perspective – most notably the route of administration, formulation attributes and food requirements for use for children with non-severe dengue. Consideration of food requirements was deemed especially important for infants and young children who are still fed with formula or breast-milk.

The group reviewed the characteristics of several paediatric dosage forms, assessing their suitability for treating both severe and non-severe dengue and considering the expected delivery models for each indication (outpatient versus inpatient care). Dose flexibility and acceptability were examined, considering the likelihood of vomiting – including persistent vomiting – and other factors influencing cultural acceptability, such as the use of suppositories. Additional considerations included stability under tropical climate conditions and gastrointestinal tolerability.

Vomiting is a common feature of dengue but varies by disease phase and clinical significance. In the early phase (the first 48–72 hours), it is often mild and manageable and does not necessarily predict disease severity. At this stage, determining which children will progress to severe dengue remains difficult (except for children with known risk factors), and early antiviral treatment – ideally in the first 2–3 days – was emphasized as critical to limit disease progression.

A key determinant for oral treatment eligibility is whether the child can retain oral fluids. Children with severe abdominal pain or persistent vomiting are unlikely to tolerate oral administration and may be entering the critical phase of disease, in which vomiting signals possible progression toward shock rather than isolated gastrointestinal upset.

In contrast, early-phase nausea and vomiting are frequently subjective and can often be managed with small, frequent fluid aliquots; forced feeding is discouraged, since it may provoke vomiting.

For children with non-severe dengue and no persistent vomiting, liquid formulations (oral liquids and syrups) were noted as offering dose flexibility and being suitable for administration in outpatient settings while also supporting the management of mild dehydration and fever. However, the group acknowledged several limitations, including their poor chemical and microbial stability – especially in tropical climates – and the frequent presence of excipients of safety concern, such as preservatives or propylene glycol, which has limited metabolic clearance among children younger than four years. In addition, these formulations are often supplied in glass bottles, which are bulky and contribute to a high supply-chain footprint.

Buccal and sublingual dosage forms were considered as a potential option for non-severe dengue during the TPP development group meetings and in the public consultation. Although the group recognized their potential usefulness for children with swallowing difficulties, it also noted that these formulations are appropriate only for low-dose (highly potent) drugs and pose a significant risk of mucosal irritation. Given the frequency of oral lesions in dengue, such irritation could negatively affect both acceptability and drug absorption. In addition, bioavailability may be compromised if the dosage form is diluted or swallowed with saliva. These routes also bypass gastrointestinal metabolism and cannot support the delivery of hydration or bulk electrolyte therapy, which may be essential in dengue management. Given all these considerations, the group concluded that buccal and sublingual formulations would not be suitable for children with non-severe dengue.

Although rectal administration could be useful for children with severe vomiting or in unconscious states, absorption is variable and acceptability differs widely, generally being lower in low- and middle-income countries.

In addition, most people with dengue who experience nausea or vomiting, including children, can be managed with small, frequent fluid aliquots, which is distinct from persistent vomiting.

For non-severe dengue, the group acknowledged feedback from the public consultation indicating a preference for oral administration, since most people with non-severe dengue are managed in outpatient settings. However, a single-dose injectable formulation was considered acceptable for therapeutic modalities that cannot be administered orally and was therefore included among the minimal TPP characteristics.

With regard to oral formulations, the group agreed that an orodispersible tablet would be the preferred dosage form (Table 2). Orodispersible tablets enable for rapid delivery of antipyretic and analgesic treatments and are especially suitable for people, including children, who have difficulty swallowing, since they disintegrate in the mouth and may facilitate administration for children experiencing vomiting. They are also advantageous in outpatient settings, where access to clean water may be limited. However, the group acknowledged that orodispersible tablets are not suitable for drugs requiring high loads of active pharmaceutical ingredient or for inherently bitter active pharmaceutical ingredients because of challenges in taste masking. In addition, they are not appropriate for high-dose rehydration or fluid and electrolyte replacement.

Overall, orodispersible tablets were considered suitable for most people with dengue who do not experience severe vomiting or who have manageable vomiting that would not compromise oral administration. Minitablets were also discussed but were not considered optimal because of acute gastrointestinal symptoms (such as nausea and vomiting) and the typical need for larger doses, which may require administering multiple tablets and increase the risk of choking for acutely ill patients.

Solid oral dosage forms that can be administered either whole or dispersed (enabling use of the same formulation for older children and adults who can swallow tablets and for younger children after dispersion in water) were considered acceptable alternatives and were therefore included among the minimal TPP characteristics, alongside injectable agents (Table 2). Although parenteral administration generally requires trained staff, hospital infrastructure and invasive procedures and is therefore not feasible for outpatient management or mild cases, the potential role of single-dose monoclonal antibodies for dengue remains under investigation. Current evidence suggests that their most likely application will be in early-stage disease rather than in severe dengue. Several promising candidates in the development pipeline are injectable monoclonal antibodies, administered either as single-dose injections or via intravenous infusion. Although oral formulations are preferred for children with non-severe dengue, injectable delivery may be acceptable and appropriate for monoclonal antibodies, given their modality and clinical use case. For this reason, single-dose injectable formulations were considered acceptable within the minimal TPP characteristics, enabling developers with effective injectable products to proceed while clearly signalling that oral formulations – ideally as orodispersible tablets – remain the preferred option for small-molecule drugs. Among injectable options, subcutaneous administration was identified as the most suitable route, given its practicality in outpatient settings and ease of administration compared with intravenous infusion.

Finally, palatability was identified as a key determinant of successful administration, since nausea makes children particularly sensitive to taste and may decrease willingness to take oral medicines; improving palatability may therefore increase the likelihood of adherence.

Consideration of investigation of milk–drug interactions in clinical development for orally administered dengue treatments for children was recommended.

Table 2. Paediatric-specific characteristics of the target product profile for dengue therapeutics discussed and agreed during the PADO for dengue meeting

Non-severe dengue		
	Minimal	Optimal
Food requirements	Requirement to be taken before or after meals, or in the absence or presence of specific foods	No specific requirements around timing of administration around food intake or co-administration with specific foods
Route	Oral or single-dose parenteral (intravenous or subcutaneous)	Oral
Formulation	Palatable solid oral dosage form that can be administered whole and dispersed, or injectable	Palatable orodispersible tablet
Severe dengue		
	Minimal	Optimal
Route	Intravenous	Intravenous and oral
Formulation	Injectable	Injectable and palatable solid oral dosage form that can be administered whole and dispersed

For severe dengue, the group acknowledged the feedback received during the public consultation, which indicated that the intravenous route is generally regarded as the most appropriate and practical route of administration, including for children, and is preferred to intramuscular administration.

However, it was also noted that an oral option, when tolerated, would be desirable even for severe dengue. Many severely ill adults classified as having severe dengue are still able to take medicines orally, thereby reducing the need for intravenous administration. Although this is more common for adults, some children may also benefit from oral formulations. For this reason, the solid oral dosage form included in the TPP should be designed to enable administration both as a whole tablet and in dispersed form.

The group emphasized that drug absorption for oral dosage forms should be specifically evaluated in clinical development studies, since people with severe dengue may experience limited mucosal absorption and impaired microcirculation, especially in the gastrointestinal tract.

Other routes of administration, such as rectal delivery via suppositories, were also proposed for consideration during the public consultation but were not considered appropriate for inclusion in the TPP, primarily because of acceptability concerns in low- and middle-income countries and high variability in drug absorption associated with this route.

Pipeline therapeutics

The group focused its discussion on therapeutics currently in Phase 2 and 3 development (Table 3) and reviewed their characteristics alongside the evidence available from completed studies (11). Mosnodenvir (formerly JNJ-1802) was retained in the exercise although its Phase 2 programme was discontinued in October 2024, since the decision reflected a strategic changing of the priorities of the company's research and development portfolio rather than any safety concerns and uncertainty about whether it will be further pursued by the same developer or others (12). Resomelagon, a small-molecule host-directed therapy, was not included, since the corresponding Phase 2 trial had only just begun enrolling participants aged 18 years and older.

Most candidates reviewed were reported to have acceptable safety profiles, with only mild and transient adverse effects, and demonstrated encouraging efficacy results. Paediatric studies (including children as young as five years) have been initiated only for SII Dengue mAb (formerly VIS513).

The group further discussed whether some of these therapeutics should be given priority for paediatric investigation, considering the existing evidence on efficacy and safety, their intended use cases and the anticipated market size for dengue therapeutics.

The group agreed that the available evidence does not enable any single therapeutic to be given priority over the others and emphasized the need to promote paediatric studies for all candidates in Phase 2 and 3 development. AV-1, EYU-688 and Fenretinide (ISLA-001) therefore added to the PADO watch list. Although the Phase 2 trial for mosnodenvir (JNJ-1802) was halted, the group considered that further investigation of this compound, including among children, would be valuable, and it was therefore also included in the PADO watch list. Given its more advanced stage of development, SII Dengue mAb (formerly VIS513) was added to the PADO priority list, which has a timeline of 3–5 years.

Table 3. Therapeutics in Phase 2 and 3 development reviewed by the PADO for dengue group

	Type of therapeutics	Mode of action	Route of administration	Current phase of development	Summary of main results	Paediatric studies
SII Dengue mAb (VIS513)	Direct-acting antiviral: monoclonal antibody	Targets an epitope on domain III of the virus' E protein that is conserved across all four serotypes	Intravenous (single dose, one- to three-hour infusion)	Phase 2 completed, Phase 3 ongoing in India (natural infection)	Strong efficacy, with significant reduction in viral load and fever duration as well as good tolerability in Phase 2	Yes (ongoing Phase 3 trial is enrolling children from five years of age)
AV-1	Direct-acting antiviral: monoclonal antibody	Targets an epitope on domain III of the virus' E protein that is conserved across all four serotypes Full mode of action not yet characterized	Intravenous (single dose, one-hour infusion)	Currently in Phase 2b (human challenge infection model)	Exhibited therapeutic effects, with significant clinical reduction of viral load and symptoms without serious adverse events	No
EYU 688	Direct-acting antiviral: small molecule	Inhibits NS4B viral protein, disrupting its interaction with the NS3 protein, which is essential for viral replication	Oral	Currently in Phase 2a (natural infection)	Preclinical studies showed pan-serotype activity, with strong efficacy in mouse models and no safety concerns	No
Fenretinide (ISLA-001)	Host-targeting antiviral: small molecule	Targets NS5 viral protein, disrupting viral replication; also shown to prevent nuclear entry of viral proteins	Oral	Phase 2a (prophylactic) completed (challenge human infection model); Phase 2b (therapeutic) started enrolment in January 2025	Results from Phase 2a showed clinically meaningful reduction in viral load and symptoms without safety concerns	No (previously studied in cancer among children)
Mosnodenvir (JNJ-1802)	Direct-acting antiviral: small molecule	Inhibits interactions between NS3 and NS4B viral proteins, blocking viral replication	Oral	Phase 2a discontinued following change of portfolio priorities by the study sponsor	Dose-dependent antiviral activity, safe and well tolerated in Phase 1 human challenge models and phase II studies	No

Research priorities

The group emphasized the need to accelerate paediatric investigations of therapeutics currently being evaluated for adults in Phase 2/3 clinical development. The importance of including children younger than two years was highlighted, alongside optimizing study design, endpoints and enrolment strategies for this age group.

The group acknowledged that the clinical presentation of dengue differs between adults and children and that these differences should be considered when developing and evaluating new therapeutics. The proportion of children younger than two years affected by dengue may vary across geographical regions and evolve over time. Participants noted that infants with severe disease were more frequently observed in earlier years, partly because of maternally derived antibody enhancement, and the disease now appears to be shifting towards older age groups.

Given this variability across regions and over time, investigation among children younger than two years was recognized as important, although they present particular challenges for recruitment and study design. Examples from other disease areas, such as monoclonal antibody development for respiratory syncytial virus, demonstrate that early inclusion of young children, sometimes in parallel with adults, is feasible and should be explored.

Comorbidities relevant to paediatric dengue studies were noted to include both obesity and malnutrition, and the importance of enrolling children and adolescents with these conditions in clinical trials was emphasized. Immune-mediated diseases (not necessarily associated with immunocompromise) were also identified as important considerations when enrolling children in relevant studies.

Conclusions and next steps

The PADO for dengue meeting brought together academic researchers, clinical experts, implementing partners and other key stakeholders involved in research and development to provide paediatric-specific feedback on the draft TPP for dengue therapeutics and to reach consensus on the first-ever PADO priority and watch lists for dengue. The group emphasized that paediatric investigations for all therapeutics in late-phase development should be planned and initiated as soon as sufficient adult data become available, typically when adult studies are in Phase 2.

The outcomes of the exercise will be widely disseminated through multiple engagement opportunities with regulators, industry, funders, civil society and the general public.

Finally, the WHO Department of Malaria and Neglected Tropical Diseases will use existing mechanisms to follow up on some of the technical discussions in the areas given priority, including identified research priorities. When needed, the GAP-f network and its working groups will be leveraged to advance and accelerate the investigation, development and introduction of priority products.

PADO-dengue priority list	PADO-dengue watch list
SII Dengue mAb (VIS513)	AV-1
	EYU 688
	Fenretinide (ISLA-001)
	Mosnodenvir (JNJ-1802)

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Note: All references were accessed on 8 December 2025.

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Annex 1. Agenda

Agenda of the PADO for dengue meeting		
Welcome and introduction	Martina Penazzato and Raman Velayudhan (WHO)	12:00–12:05
Paediatric drug optimization (PADO) meeting objectives	Tiziana Masini (WHO)	12:05–12:15
WHO policies and programme perspective	Raman Velayudhan (WHO)	12:15–12:25
Stakeholder perspectives		
Perspective from affected communities	Lucy Lum (University of Malaya, Malaysia)	12:25–12:35
Market and implementation perspective	Gabriela Chaves (Drugs for Neglected Diseases initiative consultant)	12:35–12:45
Q&A	All	12:45–12:55
Target product profile for dengue therapeutics		
Update on target product profile development process	Xin Hui Chan (WHO)	12:55–13:05
Options for paediatric formulations	Smita Salunke (European Paediatric Formulation Initiative)	13:05–13:15
Structured discussion on paediatric-focused characteristics of the target product profile	Moderator: Xin Hui Chan and Martina Penazzato (WHO)	13:15–14:45
(15 min break)		
Pipeline for dengue therapeutics		
An overview of the candidates in late-phase clinical development	Andre Siqueira (Drugs for Neglected Diseases initiative)	14:45–15:00
Structured discussion on the PADO watch list	Moderator: Tiziana Masini (WHO)	15:00–15:35
Priority research questions		
Invited talk	Leo Yii Sin (National Centre for Infectious Diseases, Singapore)	15:35–15:45
Research priorities	Moderator: Martina Penazzato (WHO)	15:45–15:55
Closing	Martina Penazzato and Raman Velayudhan (WHO)	15:55–16:00

Annex 2. List of participants

Ida Ameda	United Nations Children's Fund, Switzerland
Telma Azevedo	Médecins Sans Frontières, Switzerland
Gabriela Costa Chaves	Drugs for Neglected Diseases initiative, Switzerland
Tatiana Guimarães de Noronha	University of Oxford, United Kingdom of Great Britain and Northern Ireland
Janice Lee	Drugs for Neglected Diseases initiative, Switzerland
Yee-Sin Leo	National Centre for Infectious Diseases, National Healthcare Group, Singapore
Lucy Lum	University Malaya Medical Centre, Malaysia
Tariq Madani	King Abdulaziz University, Saudi Arabia
Neelika Malavige	University of Sri Jayawardenepura, Sri Lanka
Sébastien Morin	Medicines Patent Pool, Switzerland
Isabela Ribeiro	Drugs for Neglected Diseases initiative, Switzerland
Smita Salunke	European Paediatric Formulation Initiative, United Kingdom
Andre Siqueira	Drugs for Neglected Diseases initiative, Brazil
Isadora Siqueira	Fiocruz, Brazil
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Usa Thisyakorn	Mahidol University, Thailand
Trieu Huynh Trung	Oxford University Clinical Research Unit, Viet Nam
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